

PRUDENCE ISLAND POTTER COVE, RI, USA

WMO: 997294

Lat: 41.637N

Lon: 71.339W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.3	(c) 15.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 27.1	(k) 28.9	(l) 24.2	(m) 27.3	(n) 9.5	(o) 320	(p) 0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.8	(c) 86.0	(d) N/A	(e) 83.1	(f) N/A	(g) 80.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.4	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.4	(b) 19.8	(c) 16.9	(e) DB	(f) 6.6	(g) 91.7	(h) 4.3	(i) 3.7	(j) 3.5	(k) 94.4	(l) 0.9	(m) 96.6	(n) -1.5	(o) 98.7	(p) -4.7	101.4	(4)
			(e) WB	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	30.8	32.5	38.9	49.4	58.5	67.1	74.2	72.7	66.6	56.7	46.0	38.1		
	DBStd	16.27	9.08	7.89	7.42	5.93	5.89	5.60	4.21	3.84	5.11	6.48	7.53	8.22		
	HDD50	2078	596	490	351	80	3	0	0	0	0	18	164	376		
	HDD65	5216	1060	910	810	468	216	39	1	1	42	265	571	833		
	CDD50	3074	0	0	6	63	266	513	749	704	497	226	42	8		
	CDD65	738	0	0	0	1	14	102	285	239	89	8	0	0		
	CDH74	3495	0	0	0	16	67	471	1578	1099	241	23	0	0		
	CDH80	689	0	0	0	4	12	105	368	175	22	3	0	0		
(14) Wind	WSAvg	7.5	8.1	8.6	9.2	8.5	7.3	6.1	5.4	5.4	6.5	7.9	8.4	8.4		
(15) Precipitation	PrecAvg	46.3	3.7	3.5	4.7	4.1	3.5	3.6	3.3	3.8	4.0	3.9	3.9	4.4		
	PrecMax	56.1	6.2	6.5	15.2	8.0	7.4	10.7	10.1	7.8	9.3	13.0	6.7	7.2		
	PrecMin	37.1	1.7	1.8	0.3	1.5	1.1	0.0	1.0	0.8	0.8	0.4	0.7	1.3		
	PrecStd	5.3	1.2	1.3	3.0	1.7	1.3	2.6	1.8	1.8	1.9	2.6	1.6	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	53.3	63.9	75.0	81.3	88.3	90.9	87.7	82.7	77.1	65.4	60.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	51.2	49.1	56.4	66.3	74.7	82.7	87.1	84.1	79.1	71.0	61.8	56.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	47.9	46.1	52.2	61.7	71.3	79.1	84.0	81.8	76.9	68.2	59.3	53.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	44.2	43.2	48.8	58.7	68.0	76.1	81.3	79.7	74.6	65.9	57.3	51.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		12.5	12.9	13.1	14.3	14.1	13.3	12.8	13.0	12.8	12.4	11.7	11.5		
	5% DB	MCDBR	15.7	15.4	16.7	19.4	18.3	17.8	16.0	14.7	13.9	14.1	12.8	14.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.311	0.335	0.374	0.416	0.461	0.486	0.443	0.375	0.342	0.314	0.302		
	taud		2.502	2.481	2.460	2.383	2.299	2.209	2.150	2.261	2.424	2.525	2.562	2.559		
	Ebn at Noon		271	286	290	285	273	260	252	260	272	271	265	263		
	Edh at Noon		23	28	32	37	41	45	47	41	32	26	22	21		
(44) All-Sky Solar Radiation	RadAvg		591	869	1190	1533	1697	1853	1909	1715	1396	942	644	495		
	RadStd		45	84	85	164	167	183	102	89	102	85	46	49		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171	+85	

Nomenclature: See separate page